

Project Plan Roadmap

AI Schema Builder

Project Plan & Product Roadmap

Version 1.0 | 2025

1. Project Summary

Attribute

Detail

Project Name

AI Schema Builder

Project Owner

Engineering Lead

Start Date

Week 1 (Sprint 0)

Target MVP Release

Week 6

Full Release (v1.0)

Week 12

Team Size

3–4 engineers

Tech Stack

React, Node.js, PostgreSQL, Google Gemini Claude API

2. Sprint Plan

Phase 1 — Foundation (Weeks 1–4)

Sprint

Week

Feature

Owner

Story Points

Sprint 1

Wk 1–2

Project setup, Auth (register/login/JWT)

Backend

13

Sprint 1

Wk 1–2

Database schema + migrations

Backend

8

Sprint 1

Wk 1–2

React app scaffold + routing

Frontend

8

Sprint 2

Wk 3–4

Project CRUD (create, list, update, delete)

Backend

13

Sprint 2

Wk 3–4

Table management (add, edit, delete tables)

Full Stack

13

Sprint 2

Wk 3–4

API Key management (encrypted storage)

Backend

8

Sprint 2

Wk 3–4

Project & Table UI components

Frontend

13

Phase 2 – AI Core (Weeks 5–8)

Sprint

Week

Feature

Owner

Story Points

Sprint 3

Wk 5–6

Gemini API integration + prompt engineering (Free)

Backend

21

Sprint 3

Wk 5–6

Schema generation endpoint

Backend

13

Sprint 3

Wk 5–6

Schema editor grid UI

Frontend

13

Sprint 3

Wk 5–6

Error handling + retry logic

Backend

8

Sprint 4

Wk 7–8

FK reference detection via Claude

Backend

13

Sprint 4

Wk 7–8

Reference accept/reject UI

Frontend

8

Sprint 4

Wk 7–8

Interactive ERD visualization (React Flow)

Frontend

21

Sprint 4

Wk 7–8

Schema versioning

Backend

8

Phase 3 — Polish & Launch (Weeks 9–12)

Sprint

Week

Feature

Owner

Story Points

Sprint 5

Wk 9–10

SQL DDL export (PostgreSQL/MySQL/SQLite)

Backend

13

Sprint 5

Wk 9–10

JSON schema export

Backend

5

Sprint 5

Wk 9–10

ERD PNG/SVG export

Frontend

8

Sprint 5

Wk 9–10

End-to-end testing

QA

13

Sprint 6

Wk 11–12

Team/multi-user project sharing

Full Stack

21

Sprint 6

Wk 11–12

Performance optimization + caching

Backend

8

Sprint 6

Wk 11–12

Production deployment (AWS)

DevOps

13

Sprint 6

Wk 11–12

Documentation & user guide

Team

8

3. Team & Roles

Role

Responsibilities

Count

Backend Engineer

API design, Claude integration, DB schema, auth, exports

1–2

Frontend Engineer

React UI, ERD visualization, schema editor, export UX

1

DevOps / Infra

Docker, CI/CD, AWS deployment, monitoring

0.5 (shared)

QA Engineer

E2E testing, performance testing, regression suite

0.5 (shared)

Product Owner

Requirements, backlog grooming, stakeholder updates

0.5 (shared)

4. Risks & Mitigations

Risk

Probability

Impact

Mitigation

Gemini API latency > 5s for complex schemas

Medium

High

Async generation with polling; show progress indicator

Gemini free tier rate limits hit (250 req/day, 10 RPM)

Low

Medium

Implement exponential backoff; test with lower volume

ERD becomes unreadable for 50+ tables

Medium

Medium

Auto-layout + filter/search for large projects

Schema JSON parse failures from Gemini

Low

High

Strict prompt + JSON validation + retry logic

Scope creep on team features

High

Medium

Lock Phase 3 scope before Sprint 5; defer extras to v1.1

5. Definition of Done

All acceptance criteria for the story are met

Unit tests written with > 80% coverage on new code

Code reviewed and approved by at least one other engineer

Deployed to staging and smoke-tested

No open P0 or P1 bugs related to the story

API documentation updated if new endpoints were added

6. Success Metrics

Metric

Target

Measurement Method

Schema generation success rate

> 95%

API response logging

Average generation time

< 8 seconds

P95 latency monitoring

User task completion (schema to export)

< 5 minutes

Session analytics

Weekly active users (Month 1)

> 50

User analytics

ERD accuracy (FK suggestions accepted)

> 70%

User acceptance rate in DB

End of Project Plan